University of Exeter

MTH2003

Differential Equations

On Campus Exam	Closed Book
Additional Materials: Formula sheet on page 2	**Permitted Materials:** Approved Non-Programmable Scientific Calculators Permitted (Green)

Guidelines:

-The **duration** of this paper is 2 hrs
-Word Count: No Word Count
-Write all answers in the answer booklet provided.
-Do not open the exam paper until instructed by the Invigilator.
-Use only black pen or pencil.
-Mobile and electronic devices are not allowed.
-Do not communicate with other candidates during the exam.

Exam Instructions:

Answer Section A (50%) and any TWO of the three questions in Section B (25% for each).

USEFUL FORMULAS

These can be used without proof if and where needed.

An important boundary-value problem

$$y'' + \lambda y = 0, \qquad y(0) = 0, \quad y(L) = 0$$

has solutions $\lambda_n = (n\pi/L)^2$, $\phi_n(x) = \sin(n\pi x/L)$, $n \in \{1, 2, 3, \dots\}$.

SECTION A

1. (a) Find the special and the typical solutions of the ODE

$$\frac{\mathrm{d}y}{\mathrm{d}x} = 2\sin(x)\sqrt{y}, \qquad y \geq 0.$$

Hence find one solution for each of the following initial conditions, and determine the interval on which this solution is unique: (i) $y(\pi) = 4$; (ii) $y(0) = 0$. (15)

(b) Find the general solution $y(x)$ to the following ordinary differential equation:

$$x^2\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + x\frac{\mathrm{d}y}{\mathrm{d}x} - y = 3x^{-2}, \qquad x > 0.$$

Hence find the particular solution such that $y(1/2) = 2$, $y(1) = 0$. (10)

(c) (i) Use a trial function of the form $y = \sum_{n=0}^{\infty} a_n x^n$ to find the recurrence relation for the solution of the differential equation

$$(x^2 - 1)\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 4x\frac{\mathrm{d}y}{\mathrm{d}x} + 2y = 0.$$

Write down the first three non-zero terms of the power series for each of $y_1(x)$ and $y_2(x)$, where $y_1(x)$ is a solution such that $y_1(0) = 1$, $y_1'(0) = 0$ and $y_2(x)$ is a solution such that $y_2(0) = 0$, $y_2'(0) = 1$.

(ii) Further, write down the first six terms in the power series for the solution $y_3(x)$ such that $y_3(0) = 1$, $y_3'(x) = 1$.

(iii) Write down the solution $y_3(x)$ in a closed form, i.e. explicitly rather than as a series. (15)

(d) Consider the eigenfunction problem

$$(1-x^2)\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - 3x\frac{\mathrm{d}y}{\mathrm{d}x} + \lambda y = 0,\ x \in [-1, 1]; \qquad |x(-1)| < \infty,\ |x(1)| < \infty.$$

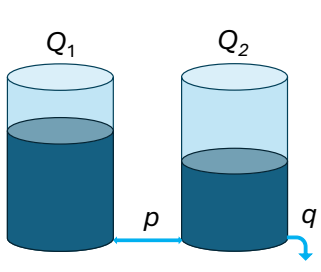
Bring this equation into the Sturm-Liouville form and hence state the orthogonality condition for the eigenfunctions of this problem. Verify that $y_* = 2x^3 - x$ is an eigenfunction of this problem and find the corresponding eigenvalue λ_*. (10)

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SECTION B

2. Two vessels are connected via a thin pipe and one of the vessels leaks. The amount of water in the first vessel is Q_1, in the second Q_2, the permittivity of the pipe between them is p, and the permittivity of the leak from the second vessel is q (see the diagram). This is described by the system of equations

(Eq2.1)
$$\begin{aligned} \frac{\mathrm{d}Q_1}{\mathrm{d}t} &= -pQ_1 + pQ_2, \\ \frac{\mathrm{d}Q_2}{\mathrm{d}t} &= pQ_1 - (p+q)Q_2. \end{aligned}$$



Let $p = 2$ and $q = 3$.

(a) Write system (Eq2.1) in the vector form $\mathrm{d}\mathbf{x}/\mathrm{d}t = \mathbf{A}\mathbf{x}$. Identify $\mathbf{x}$ and $\mathbf{A}$. Find the eigenvalues and eigenvectors of the matrix of coefficients $\mathbf{A}$. Hence write down the general solution $Q_1(t)$, $Q_2(t)$ of (Eq2.1). Write down the fundamental matrix $\mathbf{\Phi}(t)$ of this system, satisfying the initial condition $\mathbf{\Phi}(0) = \mathbf{I}$. (15)

(b) Find the solution of system (Eq2.1) for the initial conditions $Q_1(0) = 5$, $Q_2(0) = 0$. Hence find the time t_* when the amount of water in the second vessel reaches its maximum, and the value of this maximum $Q_2(t_*)$. Give both values with the precision of 3 decimal places. (5)

(c) Sketch the phase portrait of the system solved in part (a). Classify this phase portrait as one of: stable node, unstable node, stable focus, unstable focus, saddle, centre. In the same plot, highlight the trajectory corresponding to the initial-value problem solved in part (b). On this trajectory, indicate the initial point and the point of the maximum of $Q_2(t)$. (5)

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3. (a) Let $y_1(x)$ and $y_2(x)$ be two solutions of the ODE

$$\text{(Eq3.1)} \qquad \frac{\mathrm{d}^2y}{\mathrm{d}x^2} + P(x)\frac{\mathrm{d}y}{\mathrm{d}x} + Q(x)y = 0$$

on an interval $I = (\alpha, \beta)$, and let $x_0 \in I$. Show that there exists a constant $C \in \mathbb{R}$ such that for all $x \in I$, *Abel's identity* holds:

$$W(x) = C\exp\left(-\int_{x_0}^{x} P(\xi)\,\mathrm{d}\xi\right)$$

where $W(x) = y_1(x)y_2'(x) - y_2(x)y_1'(x)$. (8)

(b) Find two linearly independent solutions of the equation

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - 6\frac{\mathrm{d}y}{\mathrm{d}x} + 9y = 0.$$

Verify by direct calculation that Abel's identity holds for these two solutions. Find the corresponding constant C and hence prove that the two solutions are indeed linearly independent. (7)

(c) Suppose (Eq3.1) has solutions $y_1 = x$ and $y_2 = x^3 + 1$. Use Abel's identity to find the coefficient $P(x)$. Then find coefficient $Q(x)$. Verify by substitution that the two given solutions satisfy the resulting differential equation (Eq3.1). (10)

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4. Consider the two-dimensional Laplace equation

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \qquad \text{(Eq4.1)}$$

in a square domain, $0 \leq x \leq \pi$, $0 \leq y \leq \pi$.

(a) Using the method of separation of variables, find the solution, in the form of an infinite series, to (Eq4.1) subject to the boundary conditions $u(x,0) = 0$, $u(x,\pi) = f(x)$, $u(0,y) = 0$, $u(\pi,y) = 0$, where $f(x)$ is a given function. In particular, find the solution for $f(x) = \sin(mx)$, where m is a positive integer number. (14)

(b) Consider now the same equation (Eq4.1) with the boundary conditions $u(x,0) = 0$, $u(x,\pi) = 0$, $u(0,y) = 0$, $u(\pi,y) = \sin(my)$, where m is a positive integer number. Write down the solution to this problem using the results of part (a) (you are not expected to do a new separation of variables procedure and produce an infinite series solution). (5)

(c) Further, using the results of parts (a) and (b), find the solution of the same equation (Eq4.1) in the same domain, subject to the boundary conditions: $u(x,0) = 0$, $u(x,\pi) = \sin(2x)$, $u(0,y) = 0$, $u(\pi,y) = \sin(3y)$. (6)

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